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Adapting the traveling salesman problem to an adiabatic quantum computer

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Abstract We show how to guide a quantum computer to select an optimal tour for the traveling salesman. This is significant because it opens a rapid solution method for the wide range of applications of the traveling salesman problem, which include vehicle routing, job sequencing and data clustering.

Keywords Traveling salesman problem · Adiabatic quantum computer · Optimal tour

1 Introduction

An adiabatic quantum computer solves discrete optimization problems. It considers all solutions simultaneously and selects an optimal one very rapidly. According to [4], the time for adiabatic quantum optimization is expected to be about 4 to 6 orders of magnitude faster than for classical solvers of large problems. For example, if a problem has N inputs and takes 2^N seconds to find an exact solution, then a quantum computer can solve it in about $2^{N/2}$ seconds.

An adiabatic quantum computer employs the principle that quantum bits (qubits) can achieve an optimal state of low energy when super cooled [6]. The Ising objective function for this optimal state is

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$$\min \left(\sum_{(i,j)} s_i J_{ij} s_j + \sum_i h_i s_i \right) \tag{1}$$

where i and j are qubits,
s_i is the state of qubit i, either 0 or 1,
h_i is the energy bias for qubit i, and
J_{ij} is the coupling energy between qubits i and j.

Our goal in this paper is to transform the traveling salesman problem (TSP) into the form of the Ising function (1). This means the TSP needs to be expressed in binary variables corresponding to s in the Ising function and is limited to linear and quadratic terms corresponding to $h_i s_i$ and $s_i J_{ij} s_j$ in the Ising function (1). We will show how this can be done. The conclusion is the TSP for a large number of cities can be implemented on an adiabatic quantum processor. Current implementations are limited to 4-city TSPs due to hardware restrictions [4]. These very small TSPs have special reduction features that are not available to large TSPs.

2 The traveling salesman problem

The traveling salesman problem is a famous example in the class of NP-complete problems. A salesman needs to visit a number of locations and knows the distance between locations. The traveling salesman problem asks: what route through all locations will minimize the distance traveled? For example, a delivery person needs to drop off items at several locations. What is a shortest route that returns to the place of origin?

A traveling salesman problem (TSP) on n cities is characterized by the distances between cities. Let d_{ij} denote the distance from city i to city j for $i, j \in \{1, 2, \ldots, n\}$. If $d_{ij} = d_{ji}$ for all i and j, then the TSP is said to be *symmetric*. A *tour* for the salesman is a cyclic permutation on $\{1, 2, \ldots, n\}$, i.e., a permutation that has one cycle. The *length of tour* γ is the sum $d_{1\,\gamma(1)} + d_{2\,\gamma(2)} + \ldots + d_{n\,\gamma(n)}$. A solution for a TSP is an *optimal tour,* which is a tour ζ such that

$$\sum_{i=1}^{n} d_{i\zeta(i)} = min \sum_{i=1}^{n} d_{i\,\gamma(i)}$$

over all tours γ. References [3,8,9] provide insight about the TSP from various points of view.

3 Initial steps to adapt a directed TSP to an adiabatic quantum computer

We assume the number of cities is $n \geq 4$ and the distance d_{ij} from city i to city j is assigned for all cities i and j. We use a TSP formulation adapted from [5]. The binary variables are V_{it} representing city i occurring in position t of a tour. For example, V_{32} represents city 3 occurring as the 2nd entry of a tour, e.g., (43152) or (13524) for a 5-city TSP. The objective function is

$$\sum_{i,j=1}^{n} \sum_{t=1}^{n} d_{ij} V_{it} V_{jt+1} \quad i \neq j; \text{ subscipts } t+1 \text{ are modulo } n \tag{2}$$

We want the adiabatic quantum computer to assign 0, 1 to the variables V_{it} so that the objective function (2) is a minimum and the variables are subject to the following constraints.

$$\sum_{i=1}^{n} V_{it} = 1 \quad t = 1, 2, \ldots, n \tag{3}$$

$$\sum_{t=1}^{n} V_{it} = 1 \quad i = 1, 2, \ldots, n \tag{4}$$

Constraint (3) ensures that each position t in an outcome has exactly one city. Constraint (4) ensures that each city i occurs exactly once in an outcome.

4 Final steps to adapt a directed TSP to an adiabatic quantum computer

The Hamiltonian is a square matrix with a row and column for each variable V_{it}. The diagonal entries of the Hamiltonian are the values assigned to qubits. The off diagonal entries in the Hamiltonian are the values assigned to the connections between qubits. The entries in the Hamiltonian are the coefficients of the terms from the sum of the objective function (2) and the penalty functions derived from Constraints (3) and (4). The hardware requires a Hamiltonian to be a symmetric matrix.

If the weight of the objective function greatly exceeds the weight of the penalty functions, then the quantum solution is likely to be tilted toward a solution that satisfies the objective function but violates a constraint. On the other hand, if the weight favors the constraints, then the quantum solution is apt to find a suboptimal solution that satisfies the constraints. As in the simplex algorithm for linear programming, a lambda factor is used to balance the distances in the objective function and the coefficients in the penalty functions. See the discussion in [7, pp. 239–240], about the scalar P in Transformation 1.

A preprocessor needs to convert Constraints (3) and (4) to penalty functions only once for all TSPs with a fixed number of cities, say n_0. Then the distances for a particular TSP with n_0 cities are added to the penalty coefficients, with a lambda factor to balance them.

5 Conclusions and limitations

Adiabatic quantum computers solve discrete optimization problems. Their computational power and speed are due to their ability to quickly select a near optimal result from an exponential number of possibilities. This is an excellent machine for solving TSPs.

There is an accuracy difficulty, since [4] states "on a single program load into the quantum computer, we found the optimum 99 % of the time when we iterated 100 times." This agrees with the theoretical work in [2]. Thus, about 1 % of the time the quantum solution of a TSP may not be optimal.

Currently, an adiabatic quantum computer can handle a limited number of binary variables. This is due to binary variables equating to physical qubits in the Ising model (1). According to [5, Section 2, first paragraph], current development methods are impractical for fabricating connectivity between qubits as the number of qubits increases. This limits the size of TSPs that can be implemented.

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Appendix 1: Hardware for adiabatic quantum computing

In the adiabatic quantum computer that we used for the TSP, the quantum bits (qubits) are loops of superconducting wire, the coupling between qubits is magnetic wiring, and the housing is supercooled. Reference [6] describes this adiabatic quantum processor. It was manufactured with 128 qubits, but some have become inoperative. Fabrication limits the number of pair-wise-coupled qubits, which in turn limits the number of variables for problems that are implemented on the computer.

Appendix 2: Other features of adiabatic quantum computing

In [1] it is shown that the adiabatic quantum computing model is polynomially equivalent to the quantum circuit model which is the standard quantum computation method. This means that the TSP implementation on an adiabatic quantum computer can be transferred to the standard quantum model.

According to reference [7], many NP-complete problems have been transformed into quadratic binary problems (which the Ising model (1) requires), but the TSP is not included in the list [7, page 245].

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